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ORIGINAL ARTICLE

Peak Neutrophil-to-Lymphocyte Ratio Correlates with Clinical Outcomes in Patients with Severe Traumatic Brain Injury

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Abstract

Background: Studies suggested that the neutrophil-to-lymphocyte ratio (NLR) was associated with unfavorable outcomes in different diseases such as intracerebral hemorrhage, cardiovascular problem, cancer, and severe traumatic brain injury (sTBI). We aimed to evaluate the relationship between peak NLR and 1-year outcomes in patients with sTBI.

Methods: We retrospectively reviewed the clinical data of patients with sTBI who were treated in our department between January 2013 and January 2017. NLRs between day 1 and day 12 after admission as well as other related indicators were collected. The relationship between peak NLR and 1-year outcomes was analyzed. Factors associated with larger peak NLR were also explored.

Results: A total of 316 patients were included, and 81.3% (257/316) experienced unfavorable outcomes. Peak NLR was identified as an independent predictor for unfavorable outcomes after sTBI in multivariable logistic regression analysis (odds ratio, 1.086; 95% confidence interval, 1.037–1.137; $P < 0.001$). Its predictive value was confirmed by receiver operating characteristic analysis (area under curve = 0.775; $P < 0.001$). The day 1 NLR as well as admission Glasgow Coma Scale score was independently correlated with increased peak NLR.

Conclusion: Peak NLR was associated with the clinical prognosis after sTBI and was a promising predictor for 1-year outcomes.

Keywords: Neutrophil-to-lymphocyte ratio, Peak value, Severe traumatic brain injury, Clinical outcomes

Introduction

Despite the extensive improvement in medical care, severe traumatic brain injury (sTBI) remains as the leading cause of death worldwide [1]. According to a recent epidemiological investigation, the age-adjusted TBI mortality is 13 per 100,000 population [2], while the mortality for sTBI ranges from 30 to 50% [3–5], which mostly occurs in adolescents and young adults [6]. Victims usually survive with major disabilities, which bring substantial socioeconomic burden to their families and society.

Secondary injury resulting from edema, ischemia, oxidative stress, and inflammation is the major pathophysiology of sTBI [7]. Inflammation plays a critical role in the evolvement of neurological trauma. The imbalance between pro-inflammatory and anti-inflammatory systems largely affects the outcomes of severely injured patients [8, 9]. Roles of different inflammatory cells in TBI such as neutrophils, lymphocytes, and some other white blood cells have been thoroughly studied [10–12]. Among these cells, neutrophils and lymphocytes attract particular attention and the neutrophil-to-lymphocyte ratio (NLR) has been demonstrated to be a reliable and

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simple indicator of inflammation [13]. Moreover, the NLR is also proved to be related to the unfavorable outcomes in different diseases such as intracerebral hemorrhage [14, 15], cerebral ischemia [16, 17], tumor [18, 19], and cardiovascular problem [20]. Specifically, one study reported a significant association of the day 1 NLR after sTBI with 1-year outcomes [21]. However, the NLR was time dependent and usually increased several days after onset. Therefore, considering the day 1 NLR alone which might yield limited conclusions [22], the aim of the current study was to unravel the relationship between peak NLR and clinical outcomes in patients with sTBI. Furthermore, we assessed factors associated with larger peak NLR.

Methods

Participants

This retrospective study was based on patients admitted into our department between January 2013 and January 2017 for sTBI. Diagnosis was confirmed by head trauma history, clinical manifestations, and radiological studies. The inclusion criteria were as follows: (1) isolated head trauma; (2) an admission Glasgow Coma Scale (GCS) score less than 9; (3) age larger than 16; (4) time interval from injury to admission less than 24 h; (5) at least two consecutive NLRs over a period of 3 days or more. Missing data or loss to follow-up led to the exclusion of participants. Patients with a history of head trauma or other major diseases such as stroke, tumor, uremia, and heart failure were excluded. Approval for this study was obtained from the investigational review board of Shanghai Changzheng Hospital.

Treatment

According to the guidelines for the management of sTBI [23], the medical interventions for sTBI included head elevation, hyperventilation, sedation, steroids, seizure prophylaxis, hyperosmolar therapy, arterial pressure maintenance, early nutritional support, deep vein thrombosis prophylaxis, intracranial pressure monitoring. Tracheal intubation and mechanical ventilation were adopted based on patients' consciousness and arterial oxygen saturation. When mass effect or hernia from intracranial hematoma and brain edema was confirmed, surgical interventions would be conducted [24]. Generally, craniotomy was used for the evacuation of mass lesions, and decompressive craniectomy would be performed if there was significant or refractory intracranial hypertension [25].

Data Collection

Collected data included age, gender, time interval from injury to admission, admission GCS score, injury mechanisms, vital signs (body temperature, heart rate), blood glucose level, surgery in the first 24 h or not, length of hospital stay, 1-year Glasgow Outcome Scale (GOS) score. Neutrophil and lymphocyte accounting at different time points between day 1 and day 12 (day of admission was marked as day 1) were also collected based on the peripheral hemogram which was evaluated using venous blood samples by an automated blood counter (XN-10, Sysmex Inc., Japan). For the day 1 NLR, venous blood was obtained upon admission. For the NLRs on other days, blood samples were usually collected early in the morning. The NLR was calculated as the ratio of neutrophil count to lymphocyte count, and peak NLR was defined as the highest NLR during these 12 days. Unfavorable outcome was defined as 1-year GOS score of less than 4.

Statistical Analysis

We performed the statistical analysis with SPSS (Version 22.0; IBM, Inc., Chicago, Illinois), and the significance level was set at 0.05. Data distribution was evaluated by histograms and Kolmogorov–Smirnov test. Continuous data were either expressed as mean ± standard deviation (SD, statistical analysis: Student's t test) or median and interquartile range (IQR, statistical analysis: Mann–Whitney U test). Categorical variables were analyzed by Pearson Chi-squared or Fisher exact test. The multicollinearity was evaluated among predictive factors using the variance inflation factor (VIF), and a VIF lower than 5 indicated no significant collinearity. To identify predictors independently correlated with 1-year outcomes, we conducted a multivariate logistic regression model analysis (Method = Forward: LR) with measures showing a statistical trend ($P < 0.1$) in the univariate analysis. Receiver operating characteristic (ROC) curve analysis was performed, in which peak NLR was used as a variable and unfavorable outcome as the classification variable. Optimized peak NLR was determined by the calculation of Youden index [26, 27]. ROC curve analysis was also conducted for admission GCS score to compare its performance with peak NLR. Moreover, according to the median value of peak NLR for the entire cohort, patients were dichotomized into two groups (value smaller and larger than the median value of peak NLR). Similar multivariate logistic regression model analysis was performed to identify factors independently associated with larger peak NLR.

Results

A total of 316 patients were included in this study, with 256 (81.0%) being male. The median age for participants was 56 years (IQR, 43–63 years). Median time interval from head injury to admission was 5 h (IQR, 3–9 h). The injury was mainly caused by motor vehicle collisions

(149, 47.2%), followed by falls (141, 44.6%), assaults (17, 5.4%), and others (9, 2.8%). The median admission GCS score was 7 (IQR, 5–8), and 289 (91.5%) patients underwent surgery in the first 24 h. A total of 167 (52.8%) patients had craniotomy for the evacuation of hematoma, and 130 (41.1%) received decompressive craniectomy for their refractory or increased intracranial hypertension. On their 1-year follow-up, 257 (81.3%) patients had a GOS score of less than 4. Clinical data were compared for patients according to 1-year outcomes (Table 1). In the univariate analysis, there were significant differences for the age, admission GCS score, blood glucose level, surgery in the first 24 h, length of hospital stay, day 1 and peak NLR level between patients with favorable and unfavorable outcomes (Table 1).

In the regression models, all VIFs were lower than 2 and it proved no evident collinearity in the predictive variables. After adjusting for confounders in the multivariate logistic model, age, admission GCS score, surgery in the first 24 h, length of hospital stay, and peak NLR remained to be significantly associated with unfavorable outcomes (Table 2). ROC analysis confirmed the value of peak NLR regarding unfavorable outcomes (area under curve [AUC], 0.775; 95% confidence interval (CI), 0.698–0.852; $P<0.001$). An optimized peak NLR cutoff value of 18.16 was identified with a sensitivity of 74.3% and specificity of 72.9% (Youden Index = 0.472, Fig. 1). The admission GCS score had an AUC of 0.706 (95% CI, 0.642–0.769; $P<0.001$, Fig. 1). Median NLR for each day is shown in Fig. 2. NLR peaked frequently during day 2 and day 4, and patients with favorable outcomes had a lower NLR compared to those with unfavorable outcomes (Fig. 2).

Table 2 Multivariate analysis for predictors associated with unfavorable outcome after sTBI

Predictor	OR	95% CI	*P* value
Age (years)	1.058	1.034–1.083	< 0.001
Admission GCS score	0.629	0.452–0.876	0.006
Surgery in the first 24 h	3.460	1.206–9.927	0.021
In-hospital length (days)	1.059	1.022–1.097	0.002
Peak NLR	1.086	1.037–1.137	< 0.001

CI confidence interval; *GCS* Glasgow Coma Scale; *NLR* neutrophil-to-lymphocyte ratio; *OR* odds ratio; *sTBI* severe traumatic brain injury

Table 1 Comparison of clinical characteristics of included patients with favorable and unfavorable outcomes

Characteristic	Total (*N* = 316)	Patients with unfavorable outcome (*N* = 257)	Patients with favorable outcome (*N* = 59)	*P* value
Age (years), median (IQR)	56 (43–63)	59 (46–66)	45 (35–55)	< 0.001
Male sex, *n* (%)	256 (81.0)	209 (81.3)	47 (79.7)	0.769
Admission time (hours), median (IQR)	5 (3–9)	5 (3–9)	6 (4–9)	0.135
Injury mechanisms, *n* (%)				
Motor vehicle collisions	149 (47.2)	121 (47.1)	28 (47.5)	0.958
Falls	141 (44.6)	112 (43.6)	29 (49.2)	0.437
Assaults	17 (5.4)	16 (3.1)	1 (1.7)	0.555
Others/unknown	9 (2.8)	8 (6.2)	1 (1.7)	0.164
Admission GCS score, median (IQR)	7 (5–8)	7 (5–8)	8 (7–8)	< 0.001
Systolic pressure (mmHg), median (IQR)	136 (123–150)	135 (122–150)	140 (126–150)	0.434
Diastolic pressure (mmHg), median (IQR)	79 (70–88)	78 (69.5–88)	79 (71–89)	0.187
Body temperature (℃), median (IQR)	36.8 (36.6–37.2)	36.9 (36.6–37.2)	36.8 (36.6–37.1)	0.453
Heart rate, median (IQR)	89.5 (79–109)	89 (79–108)	90 (75–112)	0.990
Blood glucose level (mmol/L), median (IQR)	8.7 (7.4–10.38)	8.9 (7.7–10.6)	8.2 (6.9–9.8)	0.006
Surgery in the first 24 h, *n* (%)	289 (91.5)	242 (94.2)	47 (79.7)	< 0.001
Surgical interventions, *n* (%)				
Craniotomy	167 (52.8)	137 (53.3)	30 (50.8)	0.773
Decompressive craniectomy	130 (41.1)	107 (41.6)	23 (39.0)	0.709
Others	19 (6.00)	13 (5.1)	6 (10.2)	0.136
In-hospital length (days), median (IQR)	18 (12–25.75)	19 (13–27)	16 (11–20)	0.017
Day 1 NLR, median (IQR)	16.33 (11.61–20.61)	17.62 (13.08–20.89)	11.55 (8.62–14.11)	< 0.001
Peak NLR, median (IQR)	26.21 (22.2–33.34)	27.34 (23.56–35.26)	18.62 (14.33–24.44)	< 0.001

IQR interquartile range; *GCS* Glasgow Coma Scale; *N* number

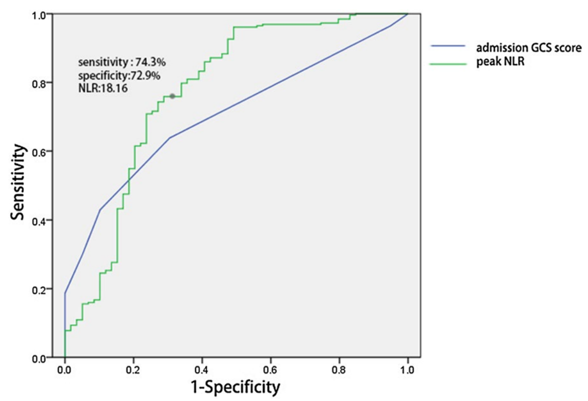

Fig. 1 ROC curve analysis for admission GCS score and peak NLR. The AUC for admission GCS score was 0.706 ($P < 0.001$). For peak NLR, the AUC was 0.775 ($P < 0.001$), and the optimized peak NLR cutoff point was 18.16. The sensitivity and specificity of it were 74.3% and 72.9%, respectively. *AUC* area under curve; *GCS* Glasgow Coma Scale; *NLR* neutrophil-to-lymphocyte ratio; *ROC* receiver operating characteristic

Considering that peak NLR was demonstrated as an independent predictor of 1-year outcomes, we investigated factors related to larger peak NLR. In the multivariate analysis, only day 1 NLR and admission GCS score were independently correlated with a larger peak NLR of more than 26.2 (median value of the entire peak NLR) (Table 3).

Discussion

In the current study, we demonstrated that the value of peak NLR was significantly higher among sTBI patients with unfavorable outcomes compared to those with favorable outcomes. Moreover, in the multivariate logistic analysis, the peak NLR was identified as an independent

Table 3 Multivariate analysis for predictors associated with a peak NLR of more than 26.2 after sTBI

Predictor	OR	95% CI	*P* value
Day 1 NLR	1.197	1.125–1.273	< 0.001
Admission GCS score	0.336	0.253–0.445	< 0.001

CI confidence interval; *GCS* Glasgow Coma Scale; *NLR* neutrophil-to-lymphocyte ratio; *OR* odds ratio; *sTBI* severe traumatic brain injury

predictor for 1-year outcomes. A cutoff value of peak NLR was identified as 18.16 with a sensitivity of 74.3% and specificity of 72.9% in predicting unfavorable outcomes. We also detected that day 1 NLR and admission GCS score were independent predictors for larger peak NLR. Our findings supported peak NLR as a promising predictor for unfavorable outcomes of sTBI.

Great efforts have been made for simple and accessible predictors of sTBI outcomes, which contribute to optimized treatment decision. The GCS score has been widely used as one of the most famous predictors of outcome after TBI [28]. In the current study, admission GCS score was identified as an independent predictor for the unfavorable outcomes with an AUC of 0.706. Similarly, the AUC for peak NLR was 0.775, demonstrating that admission GCS score and peak NLR both had a certain value and good accuracy in outcome prediction for patients of sTBI [29]. However, there are several shortcomings in the application of GCS score. Accurate assessment of GCS score is usually compromised for patients with intubation, sedation, or severe peri-orbital swelling. The slight changes in physical condition such as respiratory pattern and brain stem reflexes could not be reflected in GCS score [30], and it differentiates poorly

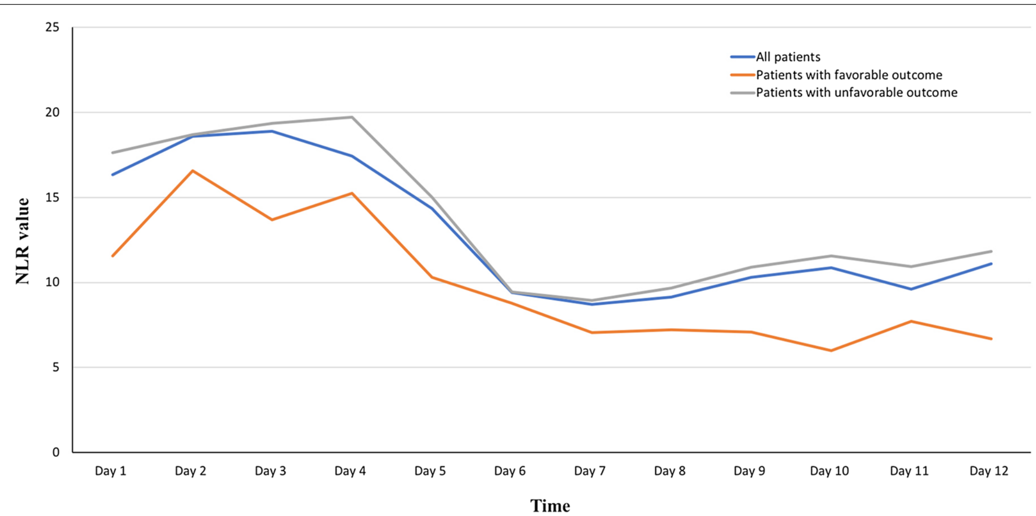

Fig. 2 Median NLR value from day 1 to day 12 between patients with different outcomes. *NLR* neutrophil-to-lymphocyte ratio

among patients with sTBI [31]. Contrastly, as an easily detectable marker, peak NLR could be obtained for almost all kinds of patients with available blood samples. Moreover, the subtle changes in a patient's neurological and physical status might be reflected in the peak NLR, leading to appropriate treatments for these patients. Therefore, peak NLR could be used as an important supplement of GCS score to improve its prognostic ability for the outcomes of sTBI patients.

NLR has been widely studied in different diseases such as the intracerebral hemorrhage, ischemia, heart disease, and cancer [14–20]. Most studies favored that NLR was a valuable prognostic marker for the outcome. Our study demonstrated the value of peak NLR in predicting 1-year unfavorable outcomes for sTBI patients for the first time. Similarly, another study explored the relationship between day 1 NLR and 1-year outcomes in sTBI [21], suggesting that increased NLR upon admission was associated with unfavorable prognosis. Yet, in our study, though the relationship between day 1 NLR and outcome was statistically significant in the univariate analysis, it became insignificant after adjusting for peak NLR in the multivariate analysis. Moreover, peak NLR was derived from NLRs of all 12 days including day 1 NLR. Our findings indicated that peak NLR might be a better predictor of outcome after sTBI with comprehensive significance.

According to our study, the NLR usually peaked during day 2 to day 4, which was consistent with another study and might correlate with featured changes of neutrophil and lymphocyte after trauma [22]. Since sTBI patients would have routine blood tests during the acute phases, obtaining peak NLR value was practicable without extra cost. It could also provide some reference values for the early interventions since therapeutic window of sTBI usually fell in the acute phase. Moreover, considering that peak NLR was a comprehensive indicator derived from multiple days, another clinical merit of the current study lied in emphasizing the importance of a dynamic observation of the NLR changes during treatment. A surge of the NLR value might respond to the patient's aggravated condition, and the aggressive interventions should therefore be considered.

The underlying mechanism for peak NLR in predicting outcomes after sTBI might be associated with the inflammatory response or immune system dysfunction. The NLR was generally regarded as a comprehensive index indicating inflammation reaction [32, 33]. The inflammatory response after TBI was not confined to the central nervous system, and an isolated brain injury would trigger complex changes in the systemic immune system. As an abundant population of circulating leukocytes, neutrophils were among the first responders in periphery after TBI. There would be a significant increase in peripheral neutrophil counts during the early 48 h after TBI [34]. Elevated neutrophils would rapidly infiltrate into brain by the choroid plexus and meningeal blood vessels [35, 36]. Neutrophils were not always neuroprotective, and they had potentials to break down the blood–brain barrier, facilitate neuronal cell death, and induce increased expression of oxidative enzymes, all of which led to deteriorated outcomes [36, 37]. Like neutrophils, lymphocytes also played a critical role in the inflammatory response. It indicated that T cell lymphocyte played a part in repairing the inflamed tissues [38]. Such function was related to the release of cytokines and growth factors, which in turn modulated microglial activation [39]. What's more, lymphocytes were the key cellular component in the cell-mediated immune system, and the decrease in lymphocytes was a sign of brain damage occurred in 12 h after injury [40]. Lymphocytopenia was linked to deteriorated immune function, and patients were prone to have complications such as fever and infection, which ultimately led to poor clinical outcomes.

Limitations

Several limitations of this study should be noted. First, this was a retrospective study, which might lead to bias caused by patient and treatment selection. Second, the blood test data during the first 2–4 days after sTBI were not available for a few patients, which would inevitably affect our results. Third, we included only patients diagnosed with sTBI in this study, whereas mild and moderate TBI patients were excluded. The sample size was relatively small, and the prognostic value of peak NLR was therefore limited. Additionally, we detected a strong association between surgery in the first 24 h and outcome (OR = 3.460), demonstrating that the outcome was largely affected by the emergency surgery. However, the relationship between surgery in the first 24 h and peak NLR was not determined in this study. Therefore, the independence of peak NLR would be challenged. More prospective studies with larger sample size were needed to validate the value of peak NLR in predicting unfavorable outcomes for sTBI patients.

Conclusion

In the present study, we detected that peak NLR was correlated with unfavorable outcomes for patients with sTBI. Peak NLR might be a promising prognostic indicator for sTBI. However, due to the limitations of the current study, more high-quality clinical investigations were needed to verify our findings and unravel the underlying mechanisms.

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Author Contribution
JC and XQ conducted the study design, data collection, and data analysis. ZL and DZ prepared the manuscript. LH reviewed and finalized the manuscript.

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Compliance with Ethical Standards

Conflict of interest
Lijun Hou received grants from National Nature Science Foundation of China. The other authors declared they had no conflicts of interest.

Ethical Approval
All procedures performed in the studies were in accordance with the ethical standards of the Changzheng Hospital and its later amendments or comparable ethical standards. For this type of study, formal consent was not required.

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